

Week#03

PROJECT 3: CI/CD PIPELINE

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Domain: DevOps Engineering

Training Program: DevOps Engineering Training

Batch: 2026

Status: Completed & Verified

1. Executive Summary

This project demonstrates the implementation of a basic **CI/CD pipeline** for a responsive web application using **GitHub and Jenkins**.

The project was designed to automate the process of retrieving source code from GitHub, validating the website files, running basic tests, and completing the deployment stage through Jenkins.

The pipeline was successfully configured and executed, with all pipeline stages completed successfully.

2. Project Objectives

The main objectives of this project were:

- Understand the basic concept of CI/CD.
- Integrate GitHub with Jenkins.
- Create a Jenkins pipeline using a `Jenkinsfile`.
- Automate the build and testing process.
- Configure multiple pipeline stages.
- Monitor pipeline execution using Jenkins Stage View.
- Verify successful CI/CD execution.

3. Technologies & Tools Used

- **Linux**
- **Git**
- **GitHub**
- **Jenkins**

- **Jenkins Pipeline**
- **HTML**
- **CSS**
- **Bash/Shell Commands**

4. Project Structure

The project contains the following important files:

DecodeLabs-Project-3-CICD-Website/

```
|  
|— index.html  
|— style.css  
|— Jenkinsfile  
|— README.md
```

5. GitHub Repository

The project source code was uploaded to GitHub and connected with Jenkins.

Repository:

<https://github.com/Salman-Ahmad542/DecodeLabs-project-3.git>

The repository contains the website source files and the Jenkins pipeline configuration.

6. Jenkins CI/CD Pipeline

A **Jenkinsfile** was created to define the CI/CD pipeline.

The pipeline contains four main stages:

Stage 1 — Checkout

The pipeline retrieves the project source code from the GitHub repository.

Stage 2 — Build

The pipeline verifies that the required website files are available, including:

index.html
style.css

Stage 3 — Test

Basic validation is performed to confirm that the expected project content exists in the HTML file.

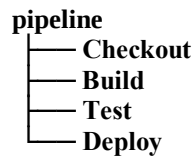
If the validation succeeds, the pipeline continues to the next stage.

Stage 4 — Deploy

The deployment stage represents the deployment process of the website.

7. Jenkinsfile

The pipeline was implemented using the following structure:

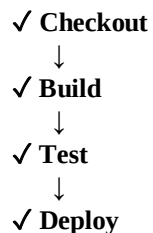


A post-build section was also configured to display a success or failure message depending on the pipeline result.

8. Jenkins Stage View

The Jenkins **Pipeline Stage View** was configured to visually monitor pipeline execution.

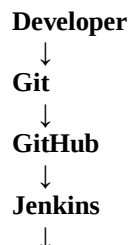
The completed pipeline shows:



All stages were executed successfully.

9. CI/CD Workflow

The complete workflow of the project is:



Checkout



Build



Test



Deploy



Successful Pipeline

This workflow demonstrates the basic concept of automating software delivery using CI/CD practices.

10. Testing & Verification

The Jenkins pipeline was executed successfully.

The following stages were verified:

Stage	Result
Checkout	✓ Successful
Build	✓ Successful
Test	✓ Successful
Deploy	✓ Successful

The Jenkins Stage View confirmed that all pipeline stages completed successfully.

11. Learning Outcomes

Through this project, the following concepts were practiced:

- Git repository management.
- GitHub integration.
- Jenkins job configuration.
- Jenkins Pipeline creation.
- **Jenkinsfile** usage.
- CI/CD workflow.
- Automated build validation.
- Basic automated testing.
- Pipeline monitoring.
- Jenkins Stage View.
- Troubleshooting Jenkins pipeline execution.

12. Challenges & Troubleshooting

During the project, different configuration and setup steps were performed to establish the GitHub-to-Jenkins workflow.

The project required configuring the correct GitHub repository, Jenkins Pipeline definition, Git branch, and **Jenkinsfile** path.

The pipeline was then tested and verified through Jenkins Stage View.

13. Final Result

The CI/CD pipeline was successfully configured and executed.

The final pipeline completed all four stages:

Checkout → Build → Test → Deploy
with successful results.

Project Status: COMPLETED & VERIFIED ✓

14. Conclusion

This project provided practical experience in implementing a basic CI/CD pipeline using GitHub and Jenkins.

The project successfully demonstrated how source code can be connected to Jenkins and processed through automated pipeline stages.

The successful execution of the pipeline and verification through Jenkins Stage View confirmed that the CI/CD workflow was configured correctly.